Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM: Simulate Hit or Miss Transformation on images.**	
Experiment No: 12	**Date:31-10-2026**	**Enrollment No:92400133121**

Aim: Simulate Hit or Miss Transformation on images.

Theory:

- The Hit or Miss transformation is a morphological operation used to detect specific patterns or shapes in an image.
- It is commonly used for shape recognition or pattern matching.
- The operation requires two structuring elements: one for foreground (hits) and another for background (misses).
- The foreground structuring element represents the desired pattern to be matched, while the background structuring element represents the complement of the pattern.
- During the transformation, the foreground structuring element is matched against the foreground pixels in the image, and the background structuring element is matched against the background pixels.
- Only those pixels that match both structuring elements are considered hits.

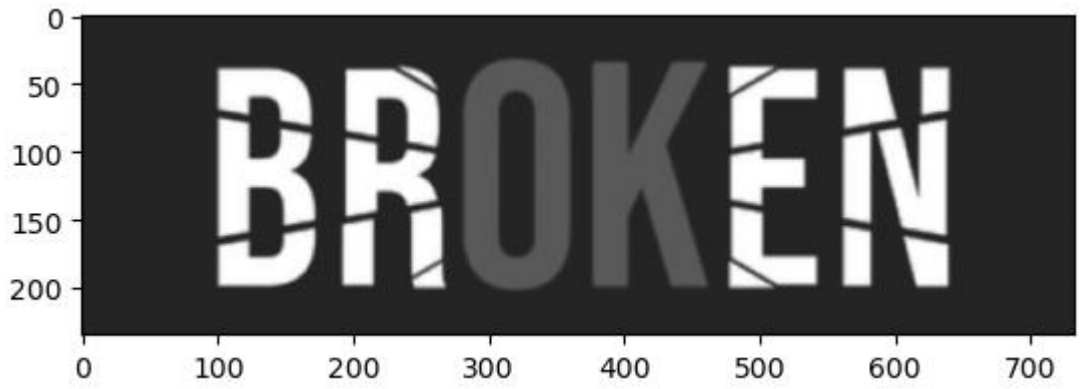
Code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

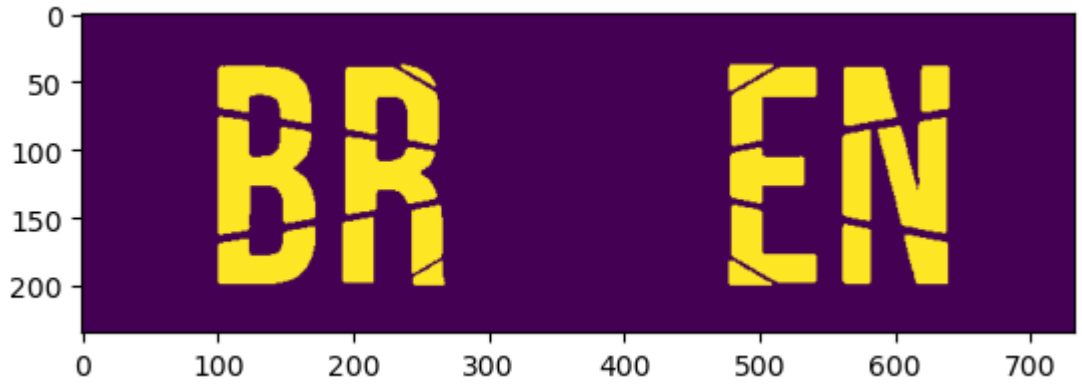
image_path = "images/BROKEN_IMAGE.png"
image = cv2.imread(image_path,0)

_,binary_image = cv2.threshold(image,123,255,cv2.THRESH_BINARY)

plt.imshow(image,cmap="gray")
```



```
plt.imshow(binary_image)
```




```
kernel = np.array([[-1, 1, -1],
                   [1, 1, 1],
                   [-1, 1, -1]],
                  dtype=np.int8)
```

→IMAGE←

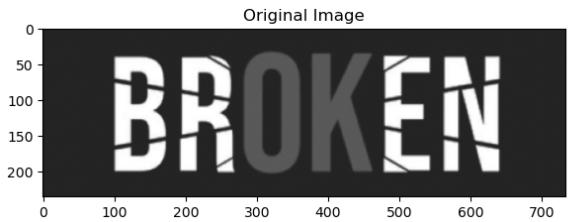
```
hitmiss_image = cv2.morphologyEx(image, cv2.MORPH_HITMISS, kernel)

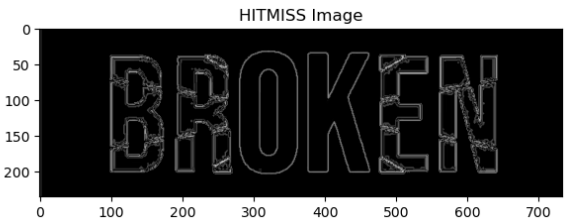
plt.figure(figsize=(15,5))

plt.subplot(1,2,1)
plt.title("Original Image")
plt.imshow(image,cmap="gray")

plt.subplot(1,2,2)
plt.title("HITMISS Image")
plt.imshow(hitmiss_image,cmap="gray")

plt.show()
```





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→BINARY_IMAGE←

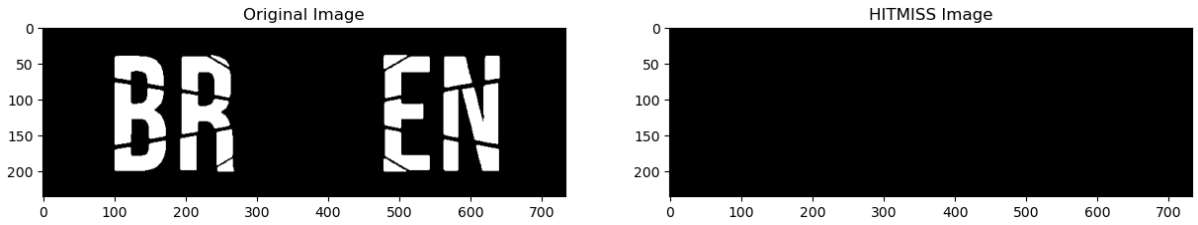
```
hitmiss_image = cv2.morphologyEx(binary_image, cv2.MORPH_HITMISS, kernel)

plt.figure(figsize=(15,5))

plt.subplot(1,2,1)
plt.title("Original Image")
plt.imshow(binary_image,cmap="gray")

plt.subplot(1,2,2)
plt.title("HITMISS Image")
plt.imshow(hitmiss_image,cmap="gray")

plt.show()
```



Conclusion:

- The Hit or Miss transformation was successfully performed.
- Specific shapes and patterns were detected using foreground and background matching.
- The structuring element was used to identify pixels matching the required pattern.
- The detected patterns were highlighted in the output image.